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Intellectual Property

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European Patent Application No: 22704518.4 - 1111
Wallac OY
Our ref: YCJ129571P.EPP

Reference is made to the Communication pursuant to Rules 161(1) and 162 EPC dated 11 August 2023 and deemed received 21 August 2023, inviting the filing of voluntary amendments and requesting the filing of a response to the International search report and written opinion/IPRP. The deadline for response is therefore 21 February 2024.

In reply, we provide the following amendments and comments.

Amendments

We enclose amended claims 1-15 to replace the claims at present on file.

A marked-up copy of the claims, to provide an indication of the basis for the claims, is also attached.

As the number of claims total 15, no claims fees are due.

It is submitted that no subject-matter has been added that extends beyond the content of the application as filed and that the amendments therefore comply with Article 123(2) EPC.

No subject-matter is abandoned. The applicant reserves the right to reintroduce deleted subject-matter into the present application and/or to pursue protection for this subject-matter by the filing of divisional applications.

Comments

Item V: Novelty

We appreciate the acknowledgment in the IPRP that the claims are novel.

Item V: Inventive Step

It was noted that the claims did not specify that the capture agent is immobilized in the well, and it was suggested that the technical problem was only solved if this was a requirement.

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The claims do now require that the capture binding agent is immobilized in the subwells.

In addition, the claimed invention is not obvious from the cited art, when read without hindsight.

The claims make it clear that a capture binding agent is immobilized in the subwells, and this binds with the target analyte, if present. Further, an upconverting nanoparticle (UCNP) labelled conjugate in the sample binds with the target analyte, if present. Thus detection of the labeled and immobilized target analyte in a respective subwell may be used to determine if the target analyte is present in the sample. A concentration of the target analyte is determined, based on a signal strength of a signal from the UCNP labeled conjugate, wherein the signal has a signal strength that corresponds to a concentration of the target analyte, if present, in the sample.

D1 is completely silent about methods of detecting a target analyte and/or performing a spatial multiplexed assay, wherein an upconverting nanoparticle (UCNP) labelled conjugate in the sample binds with the target analyte, if present. Furthermore, D1 does not appear to teach or suggest the measurement of a concentration of a target analyte.

Although UCNP labels are discussed in D2, D2 teaches towards a lateral flow assay, and does not contemplate measuring a concentration of multiple target analytes in a multiplexed array.

The skilled person is unimaginative and conservative. The skilled person does not make changes to the prior art without a motivation and an expectation of success. These are missing. There is no reason why the skilled person starting from D1 would have been motivated to make the necessary changes and choices to arrive at the claimed subject matter, with an expectation of success.

The present claimed invention provides a new and useful technical teaching that was not obvious to reach from the cited prior art.

Therefore, there is an inventive step.

Item VIII: Clarity

It was noted that PCT claims 14-23 were directed to a microplate and that the features defining this product should therefore not be method steps. The claim wording for product claims has been updated to ensure that the limitations are in respect of the product.

PCT claim 21 was considered unclear. This claim is no longer present in the claim set.

Conclusion

It is submitted that the present amendments and comments address all of the issues raised to date.

If at any stage during the examination of this application the examiner contemplates refusal of the application, Oral Proceedings are hereby requested as a precaution.

However, we would prefer that the application progresses via the usual channels of Communications under Article 94(3) EPC and/or informal telephone conversations. The examiner is also welcome to contact the undersigned by email, should this be seen as useful in progressing the application towards grant.

Yours faithfully

(signed electronically)

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